Credit Card Fraud Detection: A Data-Driven Analytical Approach

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Module: RESA C9009 - Research Process for Data Analytics

Assessment: Cross Module Project Report for Data Science

Weighting: 30%

Submission Date: 7/December/2025

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Chapter 1

Abstract

This research project shows how analyzing a credit card dataset with fraudulent transactions we can find patterns that would allow us to distinguish between fraud transactions and legitimate ones. The dataset that is being used is from the kaggle from Dal Pozzolo et al., 2021 the dataset contains 284,807 transactions from European cardholders with a 0.17% fraud transactions.

This study mainly focuses on how we can explore a dataset and use techniques to analyse the data with a CRISP-DM (Cross-Industry Standard Process for Data Mining) this is a type of framework that will be used throughout the report this will help us address the challenges of finding the fraud imbalances within the dataset.

The dataset contains a 0.17 % fraud transactions the methodologies that'll be used to find these fraudulent transactions are univariate, bivariate, and multivariate type charts for analysis along side different tests such as logistic regression, Two-sample t-test, One way ANOVA. The analyses shows a significant different in fraudulent transactions and legitimate ones, we can visualize a clustering pattern in fraudulent activities and how sampling data uncovers hidden patterns on how criminal withdraw stolen funds.

What was found about fraudulent transactions is that criminal typically steal smaller amounts and depending on the time of day there more active, a comparison between both fraud transactions and legitimate one will be shown to prove this. This study shows how important it is using effective data visualization and preprocessing data allows us to detect fraudulent transactions.

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Chapter 2

Introduction and Problem Statement

2.1 Global Context of Credit Card Fraud

Now adays using a credit card is the normal way everyone pays for stuff, cash isn't as popular as it once was the growth of credit card being used around the world incresed the amount of credit cards being stolen and fraudulent transaction also happens when researching this topic the bank of Ireland has collected data with credit card fraud showing a serious problem globally, within Ireland it was recorded to have €160 million fraudulent transactions in the year 2024 Central Bank of Ireland, 2024 This is a huge issues for everyone, businesses owner, employees ,customers anyone that uses a credit card which is mostly everyone these days with the rise in fraud is there a way for a data analysis to analyze and visualize the data of the bank dataset.

2.2 Financial and Reputational Impacts on Banks and Customers

Credit card fraud doesn't just impact our wallets there results affects businesses at a bigger scale it can ruin a companies reputation there clients could end up leaving the bank for good and move to another bank or sue the bank itself which would bring more of a financial burden to both client and business owner, enough of these issues could make the bank go bankrupt all together.

2.3 The Challenge of Imbalanced Data

The biggest problem was the challenge of detecting were the fraudulent credit cards transactions could happen since the dataset has 284,807 transactions, only 0.17% are fraudulent transactions. With such a small percentage of fraud its like finding a needle in a haystack.

2.4 Research Focus and Objectives

Within this research it will mainly focus on the foundational data analytics techniques to find and characterize patterns for fraudulent transactions. The study will cover how to explore a dataset and how to find them and visualize the fraud data transactions so we can understand the pattern thats emerging.

The objectives will be to:

1. Explore the dataset familiarize myself with it
2. Look and find a fraud transaction and identify the patterns between legit transactions from fraud.
3. Show how sampling is effective when talking about fraud.
4. Combine all the findings to have a accurate way to detect fraudulent transactions.

2.5 Dataset Selection

This dataset is ideal because it uses real world transactions that were recorded by European banks in a space of 48 hours, but to protect the peoples identity the dataset has been transformed using (PCA) which stand for principle component analysis this is done for confidentiality reasons, only time and amount was not alters from the main dataset.

2.6 CRISP-DM framework

Within this research report it follows the CRISP-DM framework structor this framework works through business understanding, data understanding, data preparation, modeling,

evaluation of findings. This approach gives us a indepth analysis while maintaining business standards for data mining practices.

Chapter 3

Literature Review

Credit card fraud is a major issue in the financial sector for data analytics because it very hard to find what transactions are fraudulent and how can you differentiate from real transactions from the other. Within this review it will talk about patterns that arise to detect fraud, imbalances in the dataset and an EDA exploratory data analysis and visualize the data to distinguish which are fraud and which aren't.

3.1 Challenge of Class Imbalance in Fraud Detection

The problem in credit card fraud analysis is distinguishing the imbalancement between real ligitamit transactions and fraudulent transactions. Dal Pozzolo et al., 2015 within this paper it mentions that fraudulent transactions usually are small which in this dataset it consist of only 0.172 percent of all transactions to be fraudulent ones it creates the scenario of "looking for a needle in a haystack".

In a research article Weiss, 1995 talks about how in traditional analytical methods often fails to detect when the data were trying to find is in the minority to address this issue the research paper by He and Garcia, 2009 tried using a technique by under sampling the dataset with the majority class, and oversampling the minority class by doing this they created a balanced dataset which revealed hidden patters.

Analyzing the fraud credit card dataset shown that when the fraud amount is in the minority under sampling and creating a visual of both classes the fraud patterns and along with the legitimate patterns hidden pattens start to show.

3.2 Patterns and Characteristics of Fraudulent Transactions

When researching about behavioral patterns the paper called Phua et al., 2010 studied the behavioral patterns of fraudulent activities by looking at indicators such as time. Within this research project time was a helpful indicator in finding the fraudulent transaction pattens which was very early mornings around 1 till 2 then a major peek at 11 am and activity dies down and on the other hand normal legal transactions starts around 8 am then till the evening night time 9 to 11 pm transaction stop people are going to sleep.

3.3 Exploring the Data and Visualizing it

In the research survey by Ahmed et al., 2016 talks about before even attempting to create complex models with dataset we first must understand its structure, detect anomalies and clean the dataset before doing anything with the dataset to increase its accuracy. If the data is note cleaned and maintained if the same dataset was feed into a powerful machine learning model it would led to misleading results, to avoid this issue the dataset must follow these steps to insure its quality of accuracy first we must check for missing values in the dataset, remove duplicate values if makes scene in most cases it makes scene but since we are analyzing a fraudulent credit card dataset we might want to keep them, this goes for detecting outliers n most cases we might want to clip the outliers but not for our scenario.

Once the data is clean its ready to be visualized, with detecting fraud this step is important because it allows us to see the emerging patterns. Baesens et al., 2021 shows how different visual technics can make data that is hard to comprehend easier to understand. within this report it uses a combination of heatmaps, boxplots, scatterplot in order to make the data easier to understand and visualize it.

3.4 Applying a CRISP-DM framework

This study with use and build upon the literature above and will do this by using the CRISP-DM framework in order to conduct visualization driven exploratory analysis of the

fraud dataset. Many of the studies mentioned about talk about predictive modeling or ai, this project will focus of the foundational steps when describing an effective model, this will be done by focusing on EDA and creating tests (t-tests, ANOVA, logistic regression). This research aim to show how classic analytical methods work and can show hidden patterns of fraudulent transaction which is supported by He and Garcia, 2009 arguing saying to have an accurately analyses and understand the data to visualize the data in an accurate way.

Chapter 4

Data Description

4.1 The Dataset Overview

The dataset that was used for this project is a publicly available one, it was taken from kaggle website the dataset is called "Credit Card Fraud Detection" Dal Pozzolo et al., 2021 this dataset contains 284,807 credit card transactions which were recorded in the span of two days on September 2013 by European cardholders. Within this dataset it contains 492 fraudulent transactions this is a total of 0.17% of the whole dataset this is a common issue in detecting fraudulent translations.

4.2 Dataset Description

This dataset contains a 31 variables there is 4 main components in the data, Time which is in second start at 0 to 172,792 second, when converted to hours its 48 hours, the second variable is the amount of money being spent in euros ranging from €0 to €25,691 class is the third variable which 0 indicates legitimate transactions and 1 showing fraudulent transactions and for the fourth variable which is the PCA component of the dataset ranging from v1 to v28 this is data transformed to comply with GDPR so it will preserve confidentiality PCA stands for Principal Component Analysis this technics still gives us the power to find fraudulent data while preserving confidentiality of the card holders. v1 shows the most significant pattern and then it dropped gradually till we reach v28 to show the least likely to show fraud data.

4.3 Data Quality

4.3.1 Completeness and Integrity

when analysis took place it show that there were no missing values in the dataset, and all variables are numrecal the large majoraty of the dataset is made up of floats (Time, Amount and v1 to v28) the class is the only variable that is an int type.

Duplicat data was found within the dataset all the duplicates were then removed i order to prevent bias results once the dataset was cleaned and checked for missing values and duplicate ones the dataset is now 283,726 transactions.

4.3.2 Transaction Distribution

When looking at the class variable there is 283,253 legitimate transactions which is (99.83%) compaired to fraudulent transactions which sits at 473 transactions this consists of (0.17%) of the whole dataset.

4.3.3 Cleaning the Dataset

There is 3 steps used to prepare the dataset, these are:

1. Loading in the dataset to checked the size of the shape and the data types it holds.
2. Removing the dublicate values there was 1,081 rows that got removed this insures quality of the dataset
3. Oulier analysis finding the inter quartile range (IQR) which was applied to the amount
    (a) Q1 (25th percentile): €5.65
    (b) Q3 (75th percentile): €77.05
    (c) IQR: €71.40
    (d) Upper Bound (Q3 + 1.5×IQR): €184.15
    (e) 8,850 transactions (3.11%) exceeded this threshold and were identified as outliers
    (f) Only 5 outlier transactions were fraudulent (0.06% of outliers)

since the dataset has a 0.17% of fraudulent transaction and the outliers comprise a 0.06% of it the outliers were not removed from the dataset if we did remove it it would damage the quality of the dataset.

4.3.4 Summary of the dataset

Table 4.1: Summary Statistics of Transaction Data

Metric	**All Transactions**	**Legitimate (Class=0)**	**Fraudulent (Class=1)**
Count	283,726	283,253 (99.83%)	473 (0.17%)
Amount (€)			
Mean	88.43	88.41	123.87
Median	22.00	22.00	9.82
Std Dev	250.38	250.38	260.21
Min	0.00	0.00	0.00
Max	25,691.16	25,691.16	2,125.87

The table shows the difference in transaction amount we can see in legitimate transaction the median and 22 and the mean is 88.43 when compared this to the fraudulent transactions we can see that the median is 9.82 which is lower the usual legit transaction but the fraudulent transaction has a greater overall mean of 123.87.

Chapter 5

Hypotheses

5.1 Hypotheses

1. H1: Does fraud transaction amount differ from legitimate ones.
2. H2: Do fraudulent transaction happen more often on certain times throughout the day.
3. H3: Can under sampling reveal hidden fraud patterns.

Chapter 6

Methodology (CRISP-DM Framework)

This study will us the cross industry standard process for data mining (CRISP-DM) framework, this is a very popular methodology which will be the project structure and wont allow it do derail. How does this methodology work by six phases which are Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.

6.0.1 Business Understanding

Within the first phase we try to convert the business question to an analytical one. How can we stop this credit card fraud from happening?

6.0.2 Data Understanding

The second phase we must collect data and go through it we must get familiar and identify if there is and quality issues.

6.0.3 Data Preparation

Preparing the dataset is a very important part if the dataset if not cleaned properly inaccuracies will most likely arise, to avoid this the dataset must be cleaned removing duplicate transactions checking to see if there no missing values across the dataset.

6.0.4 Modeling

Once the data is cleaned and ready to use we can create a visualization of the data by creating charts applying a statistical diagnosis on the dataset. For the descriptive statistic measuring the mean and median and the standard deviation comparing what we found from legitimate transactions to fraudulent ones.

Other tests that were implemented was the Two-sample t-test this test compared transaction amount with legitimate and fraudulent transactions, one way ANOVA was used to examine the different results across the transaction amount category from v1 and Logistic regression this test checks to see the relationship between PCA components and the likelihood of fraudulent transactions from accruing.

The visualization technics that were used are a combination of 3 univariate analysis which uses histograms and box plots for time and amount of money spent, bivariate analysis scatterplot were used to examine the relationship between the PCA components and the amount of money being spent, and lastly multivariate analysis a heatmap was used alongside the correlation of the PCA to check for fraud.

6.0.5 Evaluation

In this section we collected all our data, cleaned it and made sense of it and visualized it now its time to interpret our findings against our original hypotheses.

1. H1: Does fraud transaction amount differ from legitimate ones. Yes the amount is difference between fraud and legit transactions.

2. H2: Do fraudulent transaction happen more often on certain times throughout the day. Yes there is clustering fraud transaction on certain time frames.

3. H3: Can under sampling reveal hidden fraud patterns. yes it showed us where the fraud patterns were.

6.0.6 Deployment

The last step it to deploy which means translating the analytical insight into an actionable one.

Chapter 7

Data Analysis

7.1 Bivariate plot

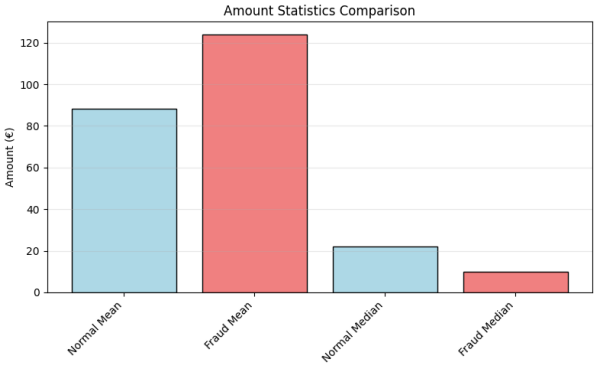


Figure 7.1: Bivariate plots comparing fraud and normal transactions

Making a chart with 2 groups the mean for the normal transactions and fraud transactions by amount of money withdrawn and did the same for the median. We can see the fraud mean is higher then the legitimate transaction because they took a greater amount of stolen money then the normal transactions but the median says the fraud transaction are lower then the normal transaction. What this shows us is that the fraudulent transactions happen in small amounts but they are consistently pulling money which then accumulates a larger mean of the total money pulled when compared to normal legitimate transaction.

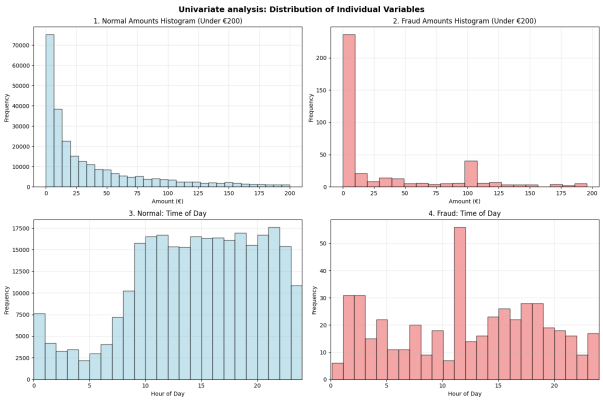


Figure 7.2: Univariate visualizations

7.2 Univariate plot

Normal transactions are in blue and fraud transactions are in red. First 2 charts check to see the different purchasing patterns of fraudulent transactions and legitimate one by taking the amount spent and how frequent it happens. The second row shows what time did the credit cards be used in a 24 hour timeline we can see that

- Normal transactions: starts around 8 am then till the evening night time 9 to 11 pm transaction stop people are going to sleep.
- Fraud transaction: We can see moderate activity in the early mornings around 1 till 2 then a major peek at 11 am.

7.3 Correlation heatmap

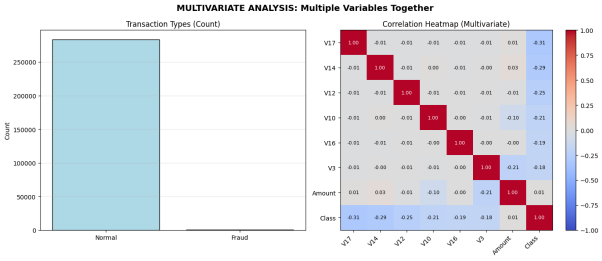


Figure 7.3: correlation heatmap

This heatmap shows how strongly related those values are in the dataset there are the highest predictors in catching fraudulent transactions

7.4 scatterplot



Figure 7.4: scatterplot

Table 7.1: Correlation Coefficients of Top Fraud Predictors

Predictor	**Correlation with Fraud (Class)**	**Interpretation**
V17	-0.313	Strong negative correlation
V14	-0.293	Strong negative correlation
V12	-0.251	Moderate negative correlation
V10	-0.207	Moderate negative correlation
V16	-0.187	Weak negative correlation
V3	-0.182	Weak negative correlation

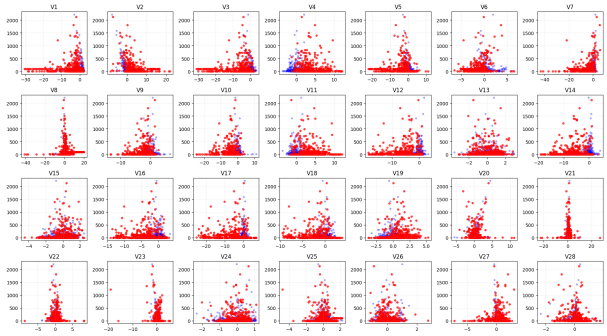


Figure 7.5: 1v to 28v scatterplot

These are all the PCA components when transformed into a scatterplot the grid shows the fraud transactions as red dots and blue transactions are normal transactions, I used a sample size of 500 for the dataset.The 2 comparisons are transaction amount with PCA values.

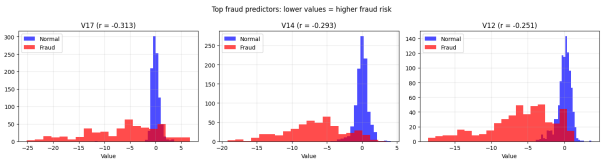


Figure 7.6: Top 3 predictors for fraud

7.5 Logistic Regression

Table 7.2: Logistic Regression Results for Fraud Prediction

Component	**Coefficient**	**Interpretation**
V14	-0.8647	Strongest fraud predictor
V12	-0.7565	Second strongest predictor
V16	-0.2465	Moderate predictor
V10	-0.1775	Weak predictor
V17	0.2167	Positive relationship (unusual)
Intercept	**-2.3721**	**Baseline fraud probability**

Table 7.3: Logistic Regression Model Performance

Metric	**Value**	**Interpretation**
Overall Accuracy	0.980 (98.0%)	Very high due to class imbalance
Fraud Detection Rate (Recall)	0.843 (84.3%)	Captures majority of fraud cases
Precision	0.092	Low due to false positives
F1-Score	0.166	Reflects precision-recall tradeoff

Predicts to see whether a transaction is fraudulent or not it does this by comparing the column called class if the value is 1 then its a fraudulent transaction if its a 0 the its a legitament one.

The top predictors that were found using correlation analysis was ('V14', 'V17', 'V12', 'V10', 'V16')

The model coefficient show the stronger the negetive number the better at predicting the chance for fraud risk to happen. The best result is v14 because it has the highest value for predicting fraud risk.

1. V14: -0.8647

2. V12: -0.7565

3. V17: 0.2167
4. V16: -0.2465
5. V10: -0.1775

- Once the log regression starts at 0 once it falls under into the negative value there is a higher risk of fraud activity to happen.
- log regression: -2.3721
- When analyzing the data I found it to be 98% right when detecting all transactions and 84.3% of detecting all fraud transactions.

What is the outcome:

- The model is not perfect it detects the large majority of the fraudulent transaction but not 100% of them it is missing 16
- v 14 and V 12 are the best at predicting, these 2 are the most important to predict the outcomes in this case.

7.6 Two-sample t-test

Table 7.4: Hypothesis Test H1: Transaction Amount Differences

Test	Two-Sample T-Test (Welch's)
Hypothesis (H1)	Fraudulent and legitimate transactions have different average amounts
Normal Mean	€88.41
Fraud Mean	€123.87
Mean Difference	€35.46 (Fraud higher)
T-statistic	-2.961
P-value	0.0032
Significance	**Significant at p ¡ 0.01**
Conclusion	**H1 Supported** - Fraud transactions have significantly higher average amounts

- This test shows whether fraudulent and normal transactions have different average amount and is there a difference in mean between each other.
- I compared the means and average of two groups using the welch t-test.

The result

- The dataset has total of 284,807 transactions legitimate ones and fraud.
- 283,253 transactions for normal transaction with a mean of €88.41
- 473 fraud transactions and the mean that was found is €123.87
- What this tells me is there is a €35.46 higher average for fraud transactions
- T-statistic: -2.961
- the negetive value tells me that the normal mean for transactions are less then fraud mean transactions.
- P-value: 0.0032170246
- If the p value falls below 0.05 which it has 0.0032 this tells us we should reject the null hypothesis, the averages are different then one another and not just luck.

7.7 One way ANOVA

Im testing to see if different transaction amounts have a different v1 value, I did this by spliting the transactions into 4 groups:

- Small (¡€50)
- Medium (€50-200)
- Large (€200-1000)
- Very Large (¿€1000)

The result is that the f-statistic = 3630.845482 This tells us that there is a huge difference in the groups when compared to there value each group is very different then one another.

For the p-value its below 0.0001 this means that there is a little to no chance of this result to be random we can confidently say there is a 99.99% confidence that each group is different.

Once we have our results we can say v1 is the primary PCA component and it shows us that small purchase have a very different v1 value then the large ones. This helps us find a pattern with the fraud transactions activity we can see in previews results that the fraud activities they take a small amounts consistently that add up to be more then your average normal transactions.

Table 7.5: One-Way ANOVA Results: V1 Differences by Transaction Amount Category

Source	**Sum of Squares**	**df**	**F-statistic**	**P-value**
Between Groups (Amount Categories)	39,721.22	3	3630.85	¡ 0.0001
Within Groups (Residual)	1,028,040.00	281,914	-	-

Table 7.6: Transaction Amount Categories for ANOVA

Category	**Range (€)**	**Sample Size**	**Description**
Small	¡ 50	192,415	Frequent, low-value transactions
Medium	50-200	69,238	Moderate everyday purchases
Large	200-1000	19,574	High-value purchases
Very Large	¿ 1000	2,499	Exceptional/luxury transactions

Chapter 8

Conclusion

within this research report it applied a data driven analytical approch using the CRISP-DM framework to guide the project goals in order to stay on track the dataset that is analysed is from a fraudulent credit card dataset with real life transactions within it, this dataset was collecteted from european card holders.

8.1 Finding

- H1: Fraudulent transactions amount of money spend is different then normal transactions.
- H2: Fraudulet transactions cluster depends on the time of day.
- H3: Under sampling of the dataset and visualizing it can reveal hidden fraud patterns.

Appendix A

Appendices

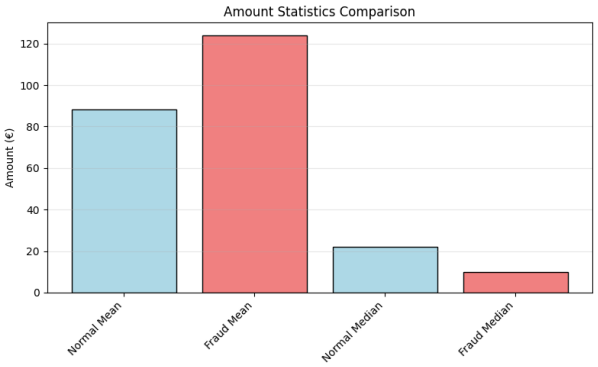


Figure A.1: Bivariate plots comparing fraud and normal transactions

A.1 Univariate plot

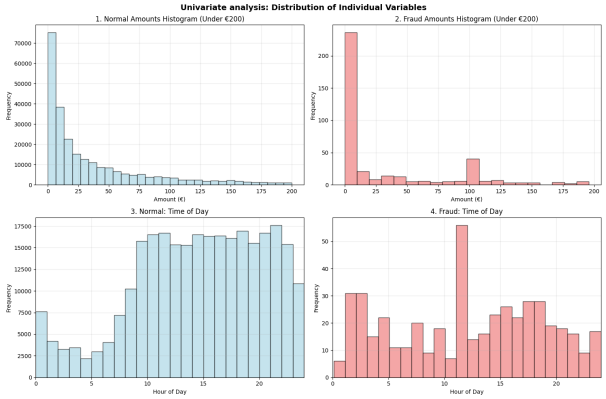


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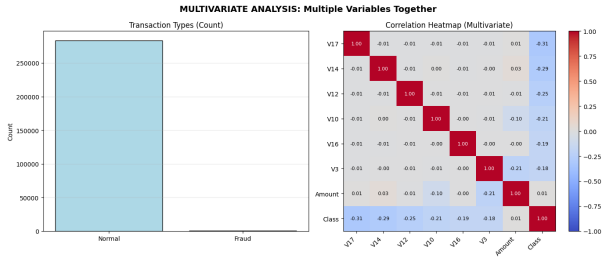


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A.3 scatterplot



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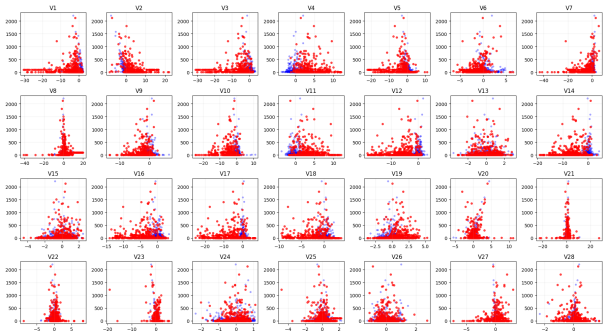


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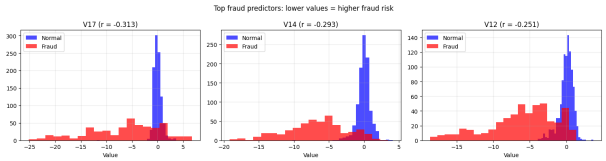


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